

<!DOCTYPE html>

```
<div>
  <span class="badge">Revision 2026</span>
  <span class="badge">Course 1004</span>
  <h1>Engineering Drawing <br>& CAD Masterclass</h1>
  <p class="subtitle">The definitive guide for Diploma students</p>
  <div style="margin-top:30px">
    <a href="#module-1" style="background:var(--blue);color:white">Module 1</a>
  </div>
</div>
<div class="card" style="background:rgba(255,255,255,0.05);border:1px solid #38bdf8">
  <h3 style="color:#38bdf8">Course Snapshot</h3>
  <ul style="list-style:none;padding:0">
    <li style="margin-bottom:12px"><strong>Credits:</strong> 10</li>
    <li style="margin-bottom:12px"><strong>Total Hours:</strong> 100</li>
    <li style="margin-bottom:12px"><strong>ESE Marks:</strong> 100</li>
    <li style="margin-bottom:12px"><strong>CIE Marks:</strong> 100</li>
  </ul>
</div>
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<h2>Module I: Geometrical Constructions & Developments</h2>

<p class="lead">Precision is the heartbeat of engineering. </p>

<div class="ml-box">

Malayalam Summary: ഈ മോഡ്യൂളിൽ പെൻസിൽ
</div>

<h3>1.1 Regular Polygons: The Pentagon</h3>

<div class="grid-2">

<div>

<p>A regular pentagon has five equal sides and internal

<ul class="step-list">

Draw the base line AB of given length (e.g., 40mm)

At point A and B, use a protractor to mark 108°.

Draw lines AC and BD of 40mm length at these angle

From C and D, swing arcs of 40mm radius to interse

Join CE and DE to complete the pentagon.

</div>

<div class="diagram-container">

<svg viewBox="0 0 200 180" width="300">

<polygon points="100,10 190,75 155,170 45,170 10,75" />

<text x="35" y="178" font-size="10">A</text><text x="190" y="75" font-size="10">B</text>

<text x="192" y="75" font-size="10">C</text><text x="45" y="178" font-size="10">D</text>

<text x="5" y="75" font-size="10">E</text>

<path d="M45,170 L55,170 A10,10 0 0 0 52,162 Z" fill="none" stroke="black" stroke-width="2"/>

<text x="55" y="165" font-size="8">108°</text>

</svg>

<p>Fig 1.1: Regular Pentagon Construction</p>

</div>

</div>

<h3>1.2 Conic Sections: The Ellipse (Concentric Circle Method)</h3>

<p>This method is highly accurate and preferred for manual c

<div class="grid-2">

<div class="diagram-container">

<svg viewBox="0 0 240 160" width="350">

<circle cx="120" cy="80" r="100" fill="none" stroke="black" stroke-width="2"/>

<circle cx="120" cy="80" r="60" fill="none" stroke="black" stroke-width="2"/>

<ellipse cx="120" cy="80" rx="100" ry="60" fill="none" stroke="black" stroke-width="2"/>

```

<line x1="20" y1="80" x2="220" y2="80" stroke="#475569"
<line x1="120" y1="20" x2="120" y2="140" stroke="#475569"
<line x1="120" y1="80" x2="170" y2="10" stroke="#cbd5e0"
<circle cx="170" cy="80" r="2" fill="#ef4444"/>
</svg>
<p><em>Fig 1.2: Concentric Circle Method</em></p>
</div>
<div>
  <ul class="step-list">
    <li>Draw two concentric circles with diameters equal to
    <li>Divide the circles into 12 equal parts (30° interval)
    <li>From each point on the <strong>outer circle</strong>
    <li>From each corresponding point on the <strong>inner circle</strong>
    <li>The intersection of these lines gives points on the
    <li>Join points with a smooth French curve.</li>
  </ul>
</div>
</div>

<div class="pro-tip">
  <strong>Pro Tip:</strong> Always keep your pencil sharp for
</div>

<h3>1.3 Development of Surfaces: Cylinder</h3>
<p>Developing a cylinder involves unfolding its curved surface
<div class="table-wrap">
  <table>
    <tr><th>Feature</th><th>Manual Calculation</th><th>CAD Software</th></tr>
    <tr><td>Length</td><td>Circumference =  $\pi \times D$ </td><td></td></tr>
    <tr><td>Height</td><td>True Height (H)</td><td>Direct 'I'</td></tr>
    <tr><td>Generators</td><td>Divide length into 12 parts</td><td></td></tr>
  </table>
</div>

```

Module II: Orthographic Projections & Sectional Views

Translating 3D reality into 2D technical data

2.1 First Angle Projection Principle

In **First Angle Projection**, the object is



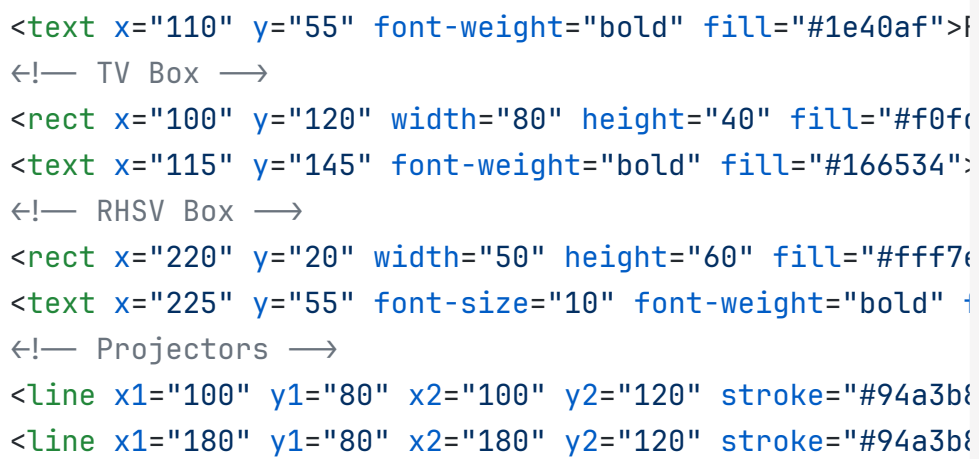
←!— Ground line →



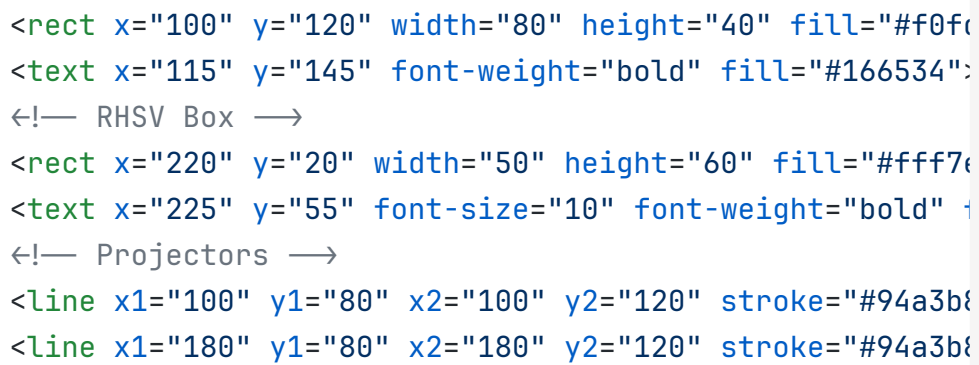
X - Y

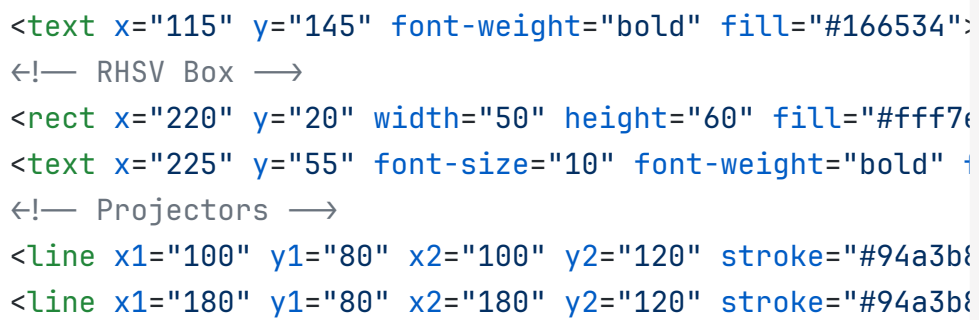
←!— FV Box →



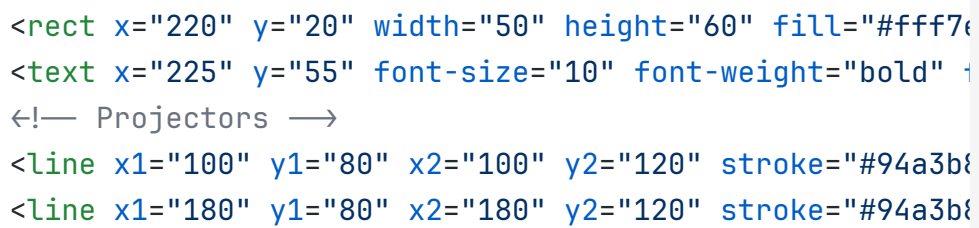


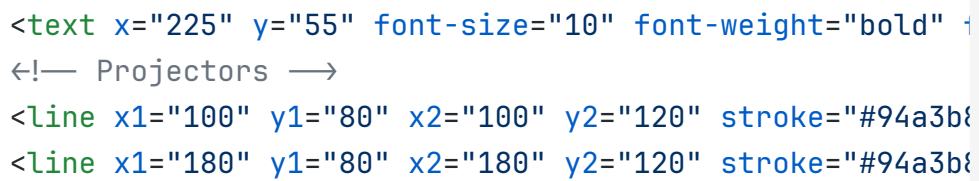
←!— TV Box →



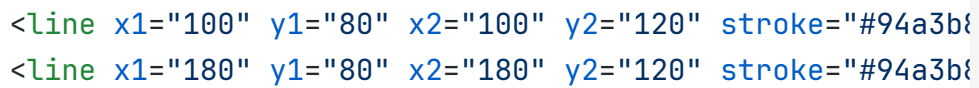


←!— RHSV Box →





←!— Projectors →





Standard Layout: FV above XY, TV below XY, Side View

2.2 Sectional Views: Revealing the Internal

A section view is used when internal hidden lines become

Hatching Standards (BIS)

Angle: 45° to the main axis.

Spacing: Uniform (usually 2mm - 3mm)

Line Type: Continuous Thin.

Rule: Never hatch through a hole



```

<rect x="20" y="20" width="110" height="60" fill="none"
<circle cx="75" cy="50" r="20" fill="#fff" stroke="#00
<!-- Hatching lines -->
<line x1="20" y1="40" x2="40" y2="20" stroke="#64748b"
<line x1="20" y1="60" x2="60" y2="20" stroke="#64748b"
<line x1="30" y1="80" x2="90" y2="20" stroke="#64748b"
<line x1="70" y1="80" x2="130" y2="20" stroke="#64748b"
<line x1="110" y1="80" x2="130" y2="60" stroke="#64748b"
</svg>
<p><em>Example: Section of a plate with a hole.</em></p>
</div>
</div>

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<h2>Module III & IV: CAD Mastery</h2>
<p class="lead">Transitioning from manual precision to digital accuracy</p>

<h3>3.1 The CAD Command Encyclopedia</h3>
<div class="table-wrap">
  <table>
    <thead>
      <tr><th>Command</th><th>Shortcut</th><th>Action</th></tr>
    </thead>
    <tbody>
      <tr><td><strong>LINE</strong></td><td><span class="cad-cmd">L</span></td><td><strong>Draw a straight line</strong></td></tr>
      <tr><td><strong>CIRCLE</strong></td><td><span class="cad-cmd">C</span></td><td><strong>Draw a circle</strong></td></tr>
      <tr><td><strong>OFFSET</strong></td><td><span class="cad-cmd">O</span></td><td><strong>Create a parallel offset</strong></td></tr>
      <tr><td><strong>TRIM</strong></td><td><span class="cad-cmd">TR</span></td><td><strong>Trim or extend lines</strong></td></tr>
      <tr><td><strong>FILLET</strong></td><td><span class="cad-cmd">F</span></td><td><strong>Round the corners of lines</strong></td></tr>
      <tr><td><strong>HATCH</strong></td><td><span class="cad-cmd">H</span></td><td><strong>Fill a closed area with a pattern</strong></td></tr>
      <tr><td><strong>EXPLODE</strong></td><td><span class="cad-cmd">EXP</span></td><td><strong>Break apart objects</strong></td></tr>
      <tr><td><strong>MIRROR</strong></td><td><span class="cad-cmd">MI</span></td><td><strong>Create a mirrored copy</strong></td></tr>
    </tbody>
  </table>
</div>

<h3>3.2 Professional CAD Workflow</h3>
<div class="grid-2">
  <div class="card" style="border-left:5px solid var(--violet); border-right:5px solid var(--violet); border-bottom:5px solid var(--violet); border-top:5px solid var(--violet); padding:10px;>
    <h4>1. Setup Phase</h4>
    <p>Before drawing a single line:</p>
    <ul>
      <li><span class="cad-cmd">UNITS</span> &rarr; Set to Imperial, Feet and Inches</li>
      <li><span class="cad-cmd">LIMITS</span> &rarr; Set to 0,0 to 1000,1000</li>
      <li><span class="cad-cmd">LAYER</span> &rarr; Create a layer named "0" (Default)</li>
      <li><span class="cad-cmd">LSPLOT</span> &rarr; Set to "On" (Plot on 0 layer)</li>
    </ul>
    <ul>
      <li><strong>Outline:</strong> Continuous, 0.40mm</li>
      <li><strong>Hidden:</strong> Dashed, 0.20mm</li>
      <li><strong>Center:</strong> Center, 0.20mm</li>
      <li><strong>Dim:</strong> Continuous, 0.15mm</li>
    </ul>
  </div>
  <div class="card" style="border-left:5px solid var(--violet); border-right:5px solid var(--violet); border-bottom:5px solid var(--violet); border-top:5px solid var(--violet); padding:10px;>
    <h4>2. Drawing Phase</h4>
    <p>Start with a rectangle, then add dimensions and hatching.</p>
    <ul>
      <li><span class="cad-cmd">RECTANG</span> &rarr; Draw a rectangle with dimensions 10x20</li>
      <li><span class="cad-cmd">DIMLINEAR</span> &rarr; Dimension the length of the rectangle</li>
      <li><span class="cad-cmd">HATCH</span> &rarr; Hatch the top half of the rectangle</li>
    </ul>
  </div>
</div>
```

```
<div class="card" style="border-left:5px solid var(--green)
  <h4>2. Drafting Phase</h4>
  <p>The "Skeleton" method:</p>
  <ol>
    <li>Draw Center Lines first.</li>
    <li>Use <span class="cad-cmd">OFFSET</span> to create
    <li>Use <span class="cad-cmd">TRIM</span> and <span class="cad-cmd">EXTEND</span>
    <li>Apply <span class="cad-cmd">HATCH</span> last for
    <li>Use <span class="cad-cmd">DIMLINEAR</span> for size dimensions.
  </ol>
</div>
</div>
```

```
<div class="pro-tip">
  <strong>CAD Power Move:</strong> Use the <strong>Object Snaps</strong> feature.
</div>
```

<h3>3.3 Sectional View Example in CAD</h3>

<p>To draw a sectional view of a hollow cylinder in CAD:</p>

```
<div class="step-list">
  <li>Draw the Front View using two concentric circles (<span class="cad-cmd">C</span>).
  <li>Draw the Top View by projecting lines downwards.</li>
  <li>Draw a horizontal cutting plane line across the FV.</li>
  <li>In the TV, identify the area cut by the plane.</li>
  <li>Command <span class="cad-cmd">H</span> &arr; Select the area to hatch.
  <li>Ensure the hatch scale is adjusted so lines are visible.
</div>
```

```
<h2>Quality Assurance & Checklist</h2>
<div class="grid-2">
  <div class="card">
    <h3>Manual Drawing Checklist</h3>
    <label><input type="checkbox"> All construction lines are light blue
    <label><input type="checkbox"> Final outlines are dark blue
    <label><input type="checkbox"> Arrowheads are 3:1 ratio
    <label><input type="checkbox"> Lettering is uniform and clear
  </div>
  <div class="card">
    <h3>CAD Drawing Checklist</h3>
    <label><input type="checkbox"> Correct layers used for each element
    <label><input type="checkbox"> No overlapping or duplicate lines
    <label><input type="checkbox"> Hatching does not cross object boundaries
    <label><input type="checkbox"> Scale is clearly mentioned
  </div>
</div>
```

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<p>Authored by Subject Experts for Course 1004: Engineering Drawing 1</p>
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